OPTIMAL DELIVERY ROUTE PLANNING FOR E-COMMERCE APPLICATIONS

A COMPARATIVE STUDY OF THE TRAVELLING SALESPERSON PROBLEM USING BI BOUND AND NEAREST NEIGHBOR ALGORITHMS

DESIGN AND ANALYSIS OF ALGORITHMS •

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WHY ROUTE OPTIMIZATION MATTERS

01 E-commerce growth increases daily delivery demand.

02 Multiple destinations make route planning complex.

03 Lower travel cost improves margins and customer satisfaction.

04 Algorithms turn routing decisions into systematic optimization.

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THE PROBLEM: ONE COMPLETE, MINIMUM-COST TOUR

Given n delivery locations and pairwise travel costs, find a route that:

CONSTRAINT 01

visits every location exactly once

CONSTRAINT 02

returns to the starting location

CONSTRAINT 03

minimizes total travel cost

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PROJECT OBJECTIVES

01 COST EFFICIENCY

Optimize practical delivery routes.

02 MINIMUM TOUR

Reduce total travel cost.

03 EXACT METHOD

Implement Branch and Bound.

04 FAST HEURISTIC

Implement Nearest Neighbor.

05 COMPARISON

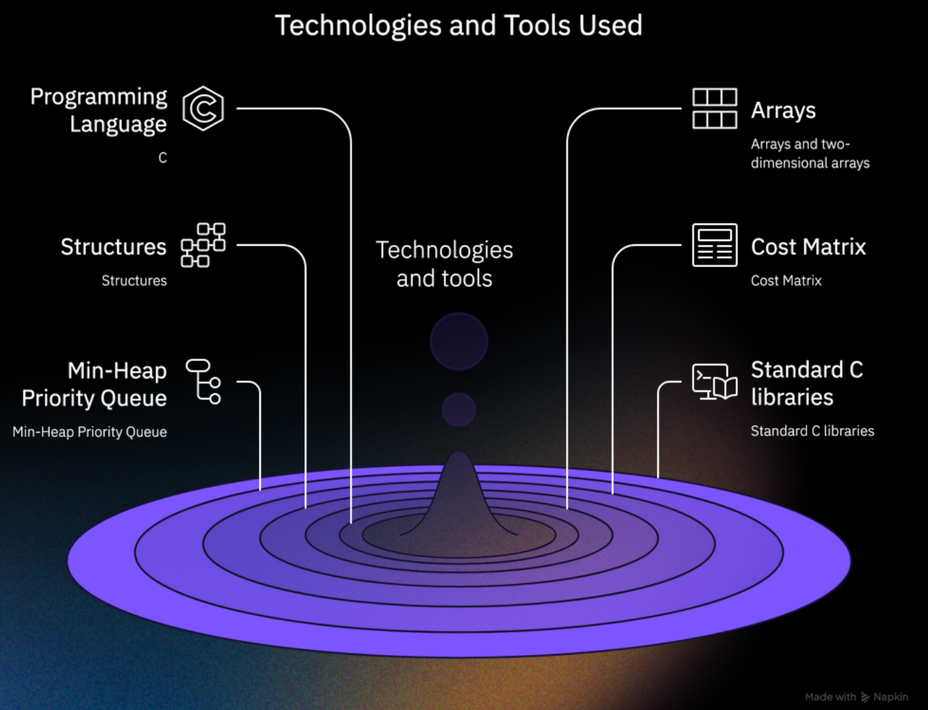
FEvaluate effectiveness and efficiency.

06 DAA PRACTICE

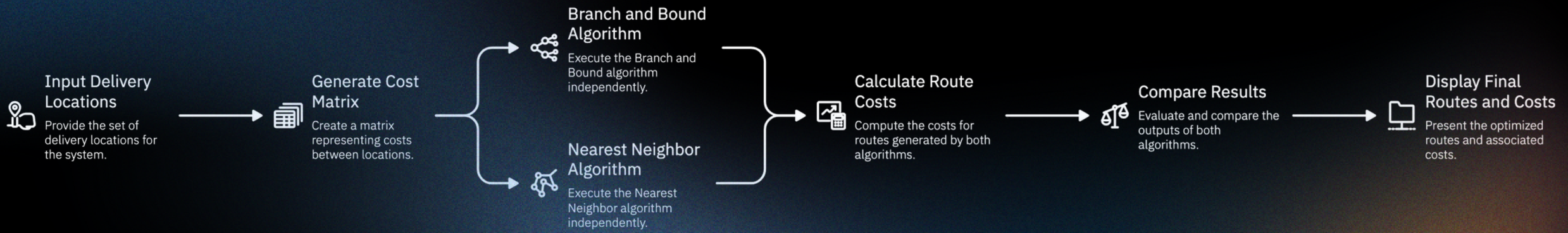
Apply algorithm design principles.

TECHNOLOGIES AND TOOLS

C - Language	Core Implementation
1D ARRAYS	Visited nodes and routes
2D MATRIX	Pairwise travel costs
STRUCTURES	Search-state organization
MIN-HEAP	Lowest-bound priority queue
STANDARD LIBRARIES	I/O and memory management



SYSTEM ARCHITECTURE



TRAVELLING SALESPERSON PROBLEM (TSP)

Find the minimum-cost Hamiltonian cycle in a weighted graph.

01 **VERTICES**

Delivery locations

02 **EDGES**

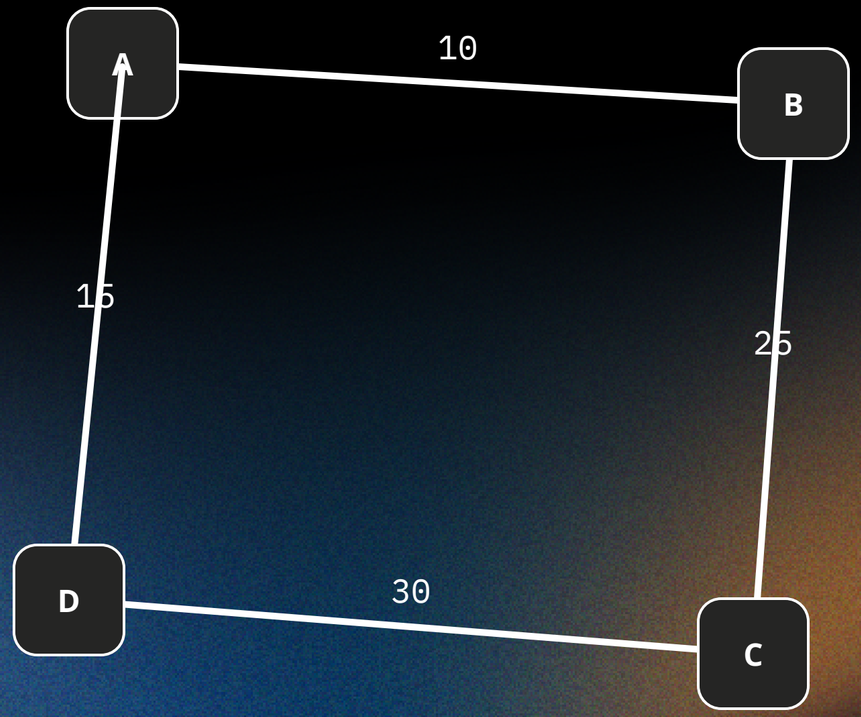
Travel connections

03 **WEIGHTS**

Travel costs

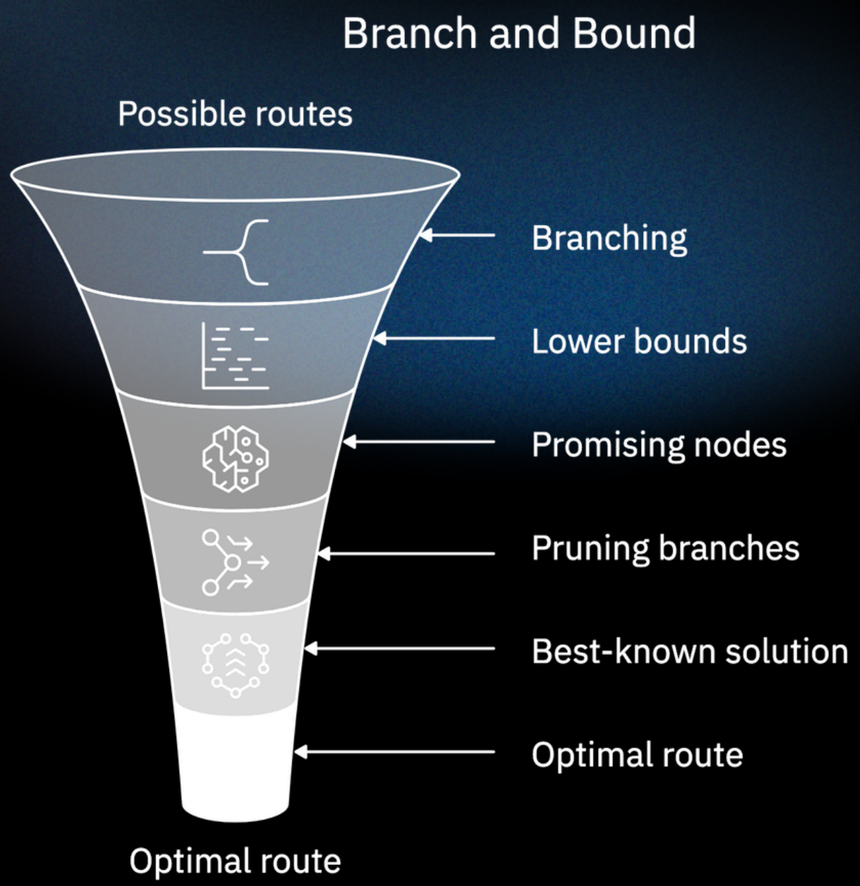
04 **GOAL**

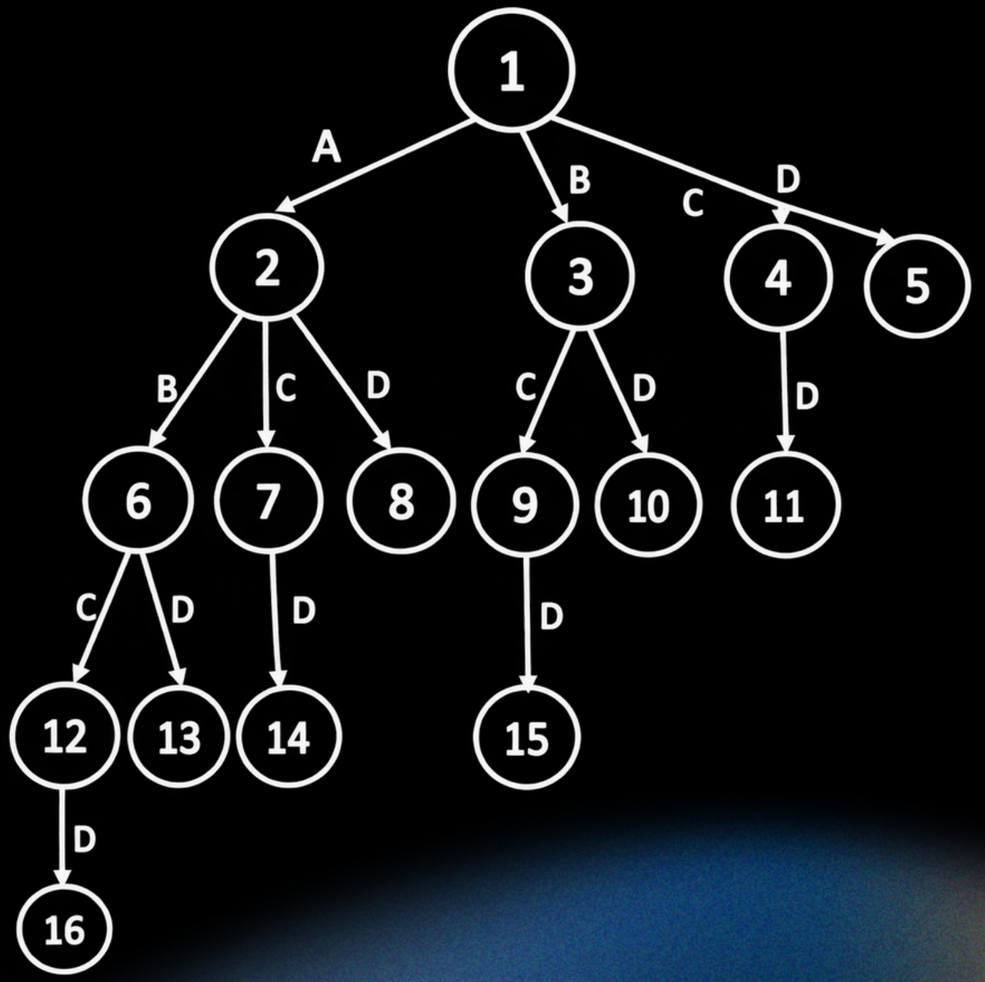
Minimum closed tour



ALGORITHM 1 — BRANCH AND BOUND

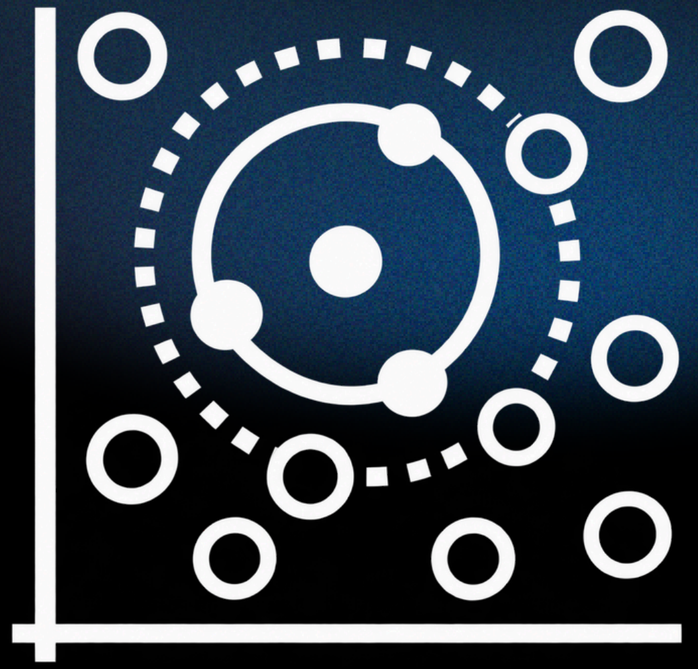
Exact search with systematic pruning





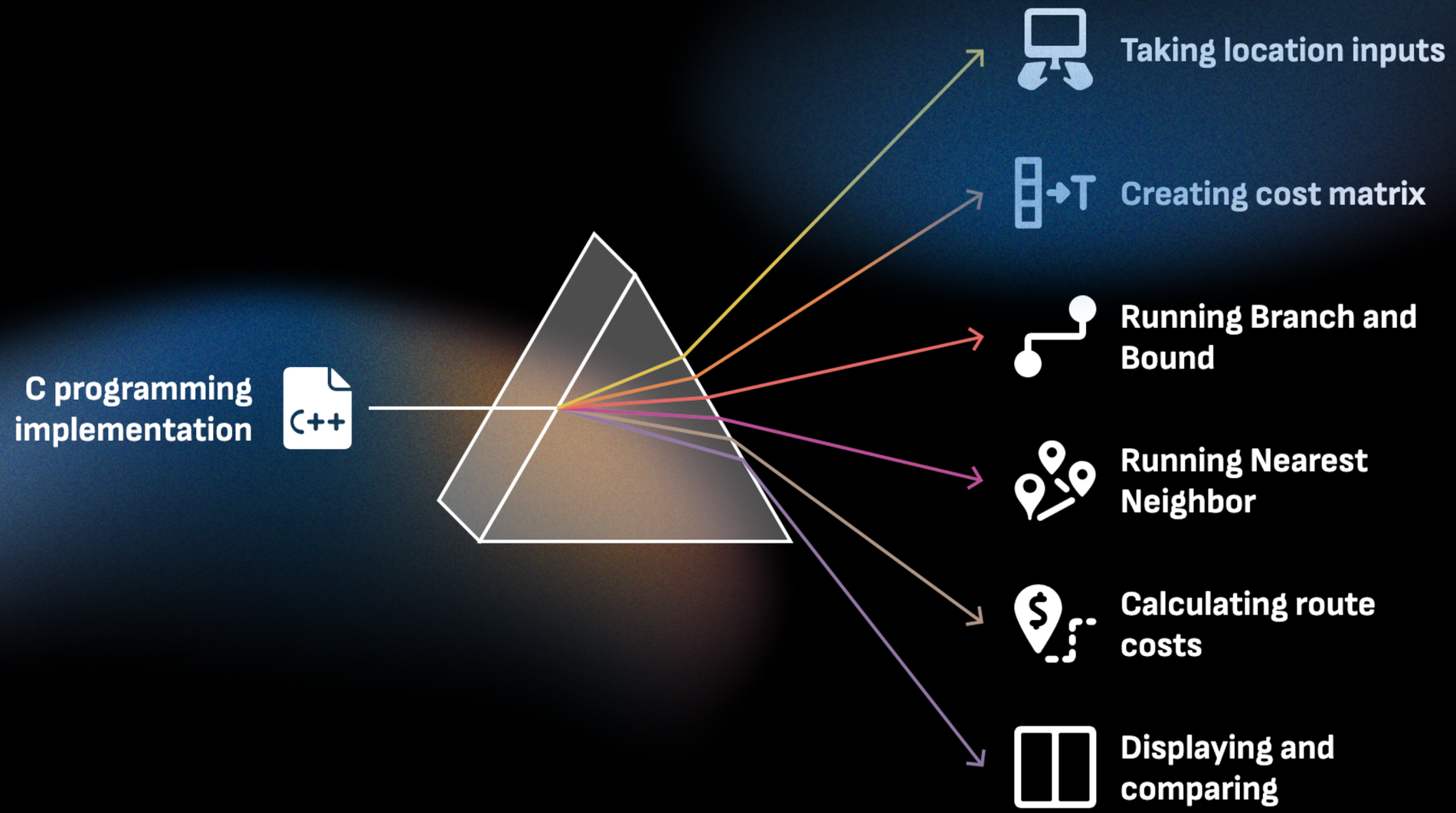
ALGORITHM 2 — NEAREST NEIGHBOR

A fast greedy heuristic



Simple • O(n²) • Fast • Not guaranteed optimal

IMPLEMENTATION AND WORKFLOW



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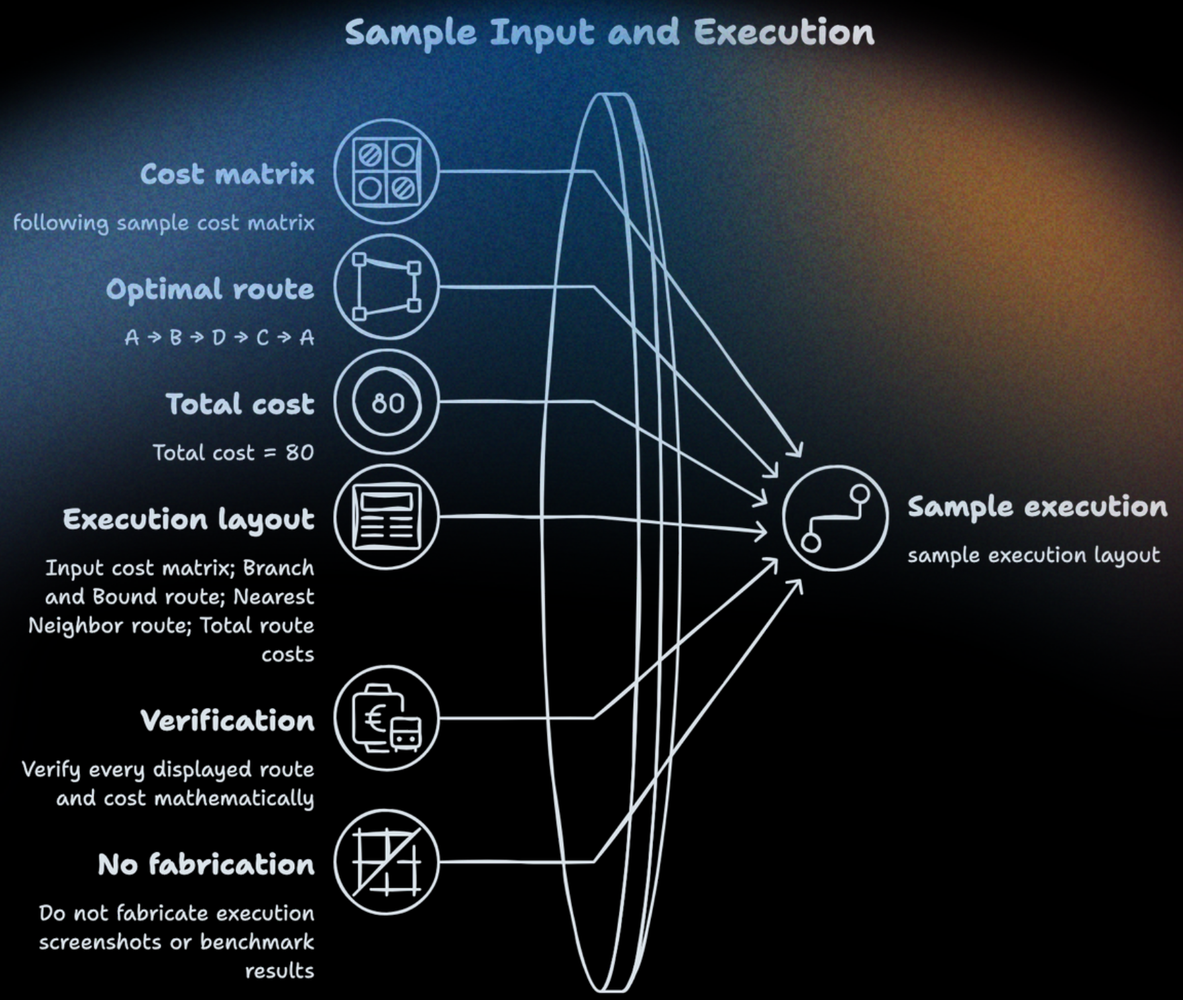
SAMPLE INPUT AND VERIFIED EXECUTION

	A	B	C	D
A	0	10	15	20
B	10	0	35	25
C	15	35	0	30
D	20	25	30	0



ALGORITHM COMPARISON AND RESULTS

CRITERION	BRANCH AND BOUND	NEAREST NEIGHBOR
Approach	Exact search	Greedy heuristic
Optimality	Guaranteed	Approximate
Time	Exponential worst-case	$O(n^2)$
Memory	Depends on heap size	O(n) visited array
Use	Exact optimization	Fast routing
Sample cost	80	80
Execution time	Not measured	Not measured

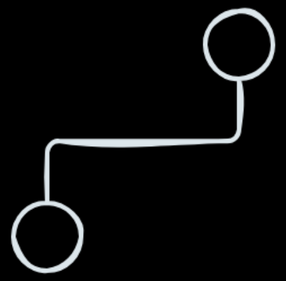
ADVANTAGES, LIMITATIONS, AND FUTURE SCOPE

Advantages

Reduces delivery route costs; demonstrates algorithmic optimization; provides exact and approximate solutions; helps understand practical DAA applications.

Limitations

Branch and Bound can become computationally expensive as locations increase; Nearest Neighbor may produce a suboptimal route; the current model uses fixed travel costs.



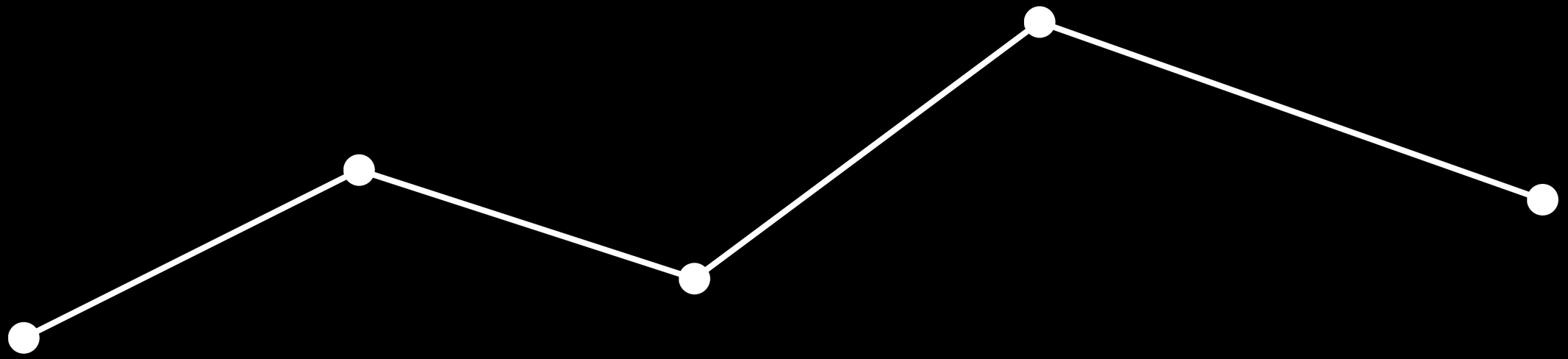
Future Scope

Real-time traffic integration; GPS-based delivery locations; dynamic route optimization; multiple delivery vehicles; time-window-based delivery planning; integration with e-commerce platforms.

THANK YOU

QUESTIONS & ANSWERS

Branch and Bound finds the optimal route. Nearest Neighbor finds a fast approximation. Together, they demonstrate how algorithm design supports practical delivery optimization.



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